



LINUX
SECURITY
SUMMIT
EUROPE

Update on Landlock

Observability, process-wide control, networking and more

Mickaël Salaün (Cloudflare)
Günther Noack (Google)

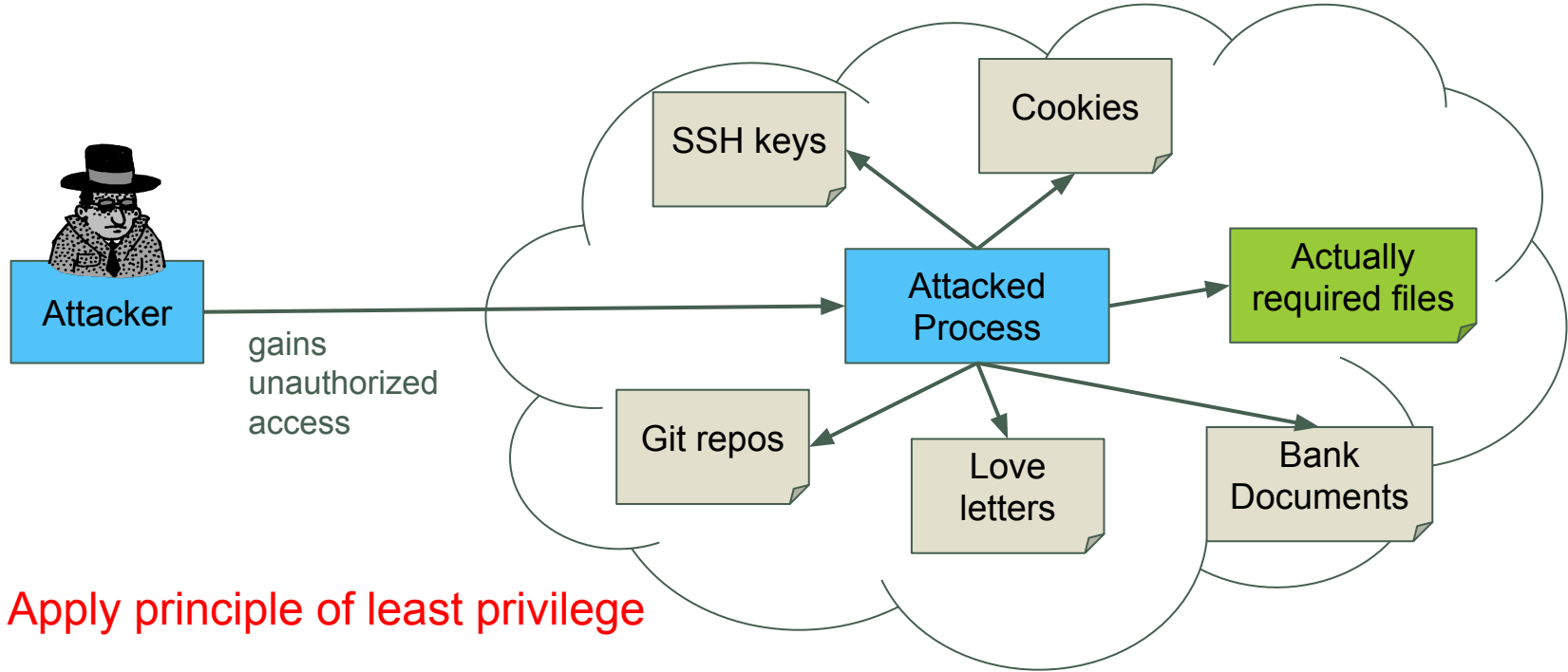
Landlock



Outline

- Landlock Motivation
- Technical Overview
- New Features in Landlock
- Upcoming Features

Motivation



An aerial photograph of a historic city, likely Prague, featuring a large Gothic cathedral with two prominent spires. The city is densely packed with buildings, and the image is overlaid with a teal gradient. The text "Technical Overview" is centered in white.

Technical Overview

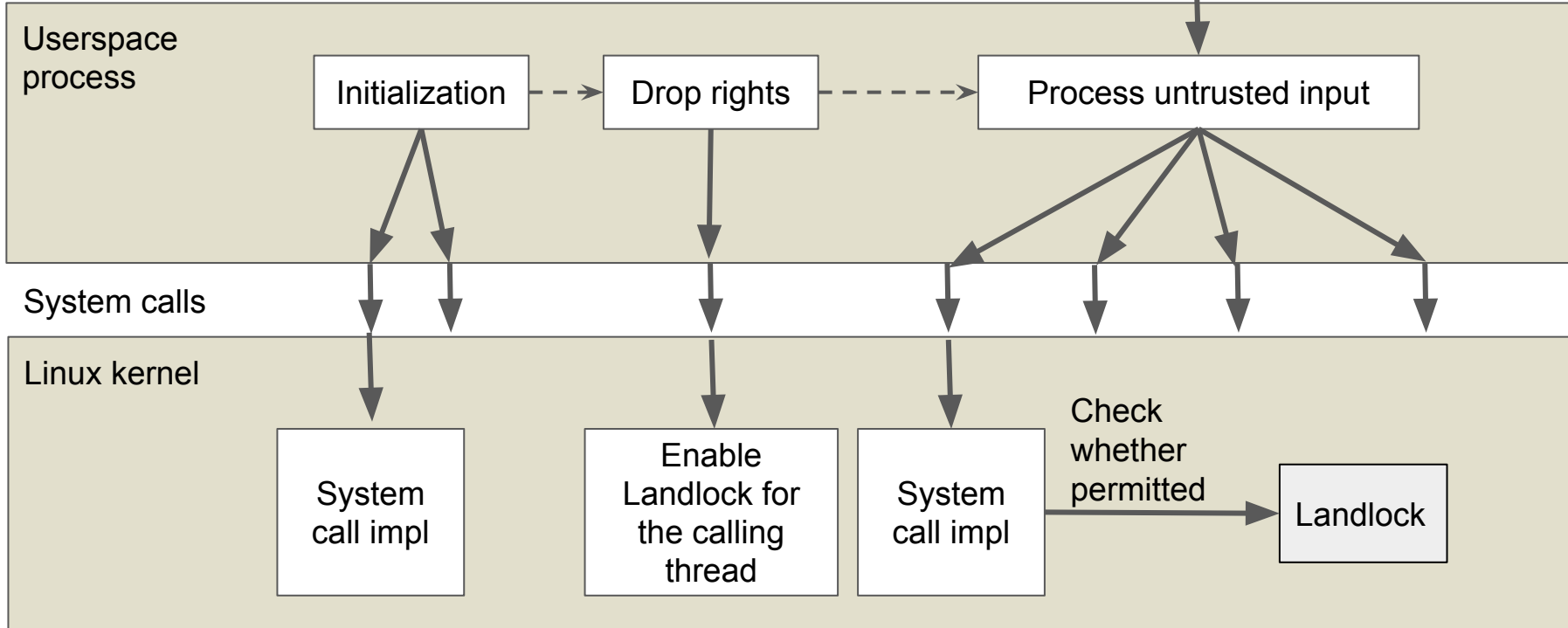
Complement solutions

- Namespaces compose kernel resources
- AppArmor, SELinux... are system-wide access control systems
- seccomp filters raw syscall arguments
- Landlock controls access to kernel resources

Demo time

<https://github.com/landlock-lsm/landlock-observability/tree/main/lltop>

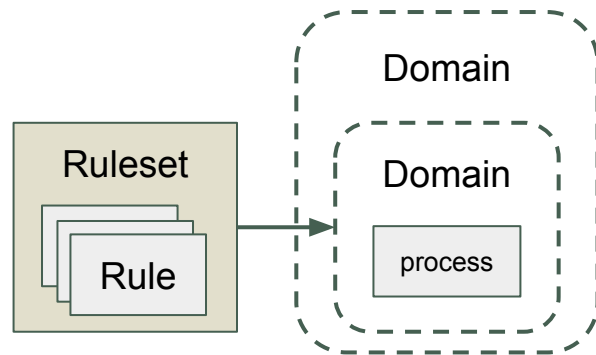
Architecture of a landlocked program



Dropping access rights

An (unprivileged) process sandboxes itself:

- `landlock_create_ruleset()`: declare what is *handled* (denied by default)
- `landlock_add_rule()`: add exceptions (paths, ports)
- `landlock_restrict_self()`: enforce policy
- Restrictions can only get tighter
- Filesystem rules apply to directory hierarchies



Restrictable access rights as of mid-2024

LANDLOCK_ACCESS_FS_*:

- Opening files for reading and writing (and truncation)
- Reading directories
- Mutating directories: Creation of different file types, removal of files and directories
- IOCTL on device files

LANDLOCK_ACCESS_NET_*:

- ..._BIND_TCP, ..._CONNECT_TCP: bind() and connect().

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New Features

Incremental development

Landlock is an unprivileged access control system which is incrementally gaining new ways to control sandboxed processes.

Each new feature is tied to an ABI version, available to user space, which makes it possible to follow a best-effort security approach to protect users as much as possible according to their kernel version: backward and forward compatibility.

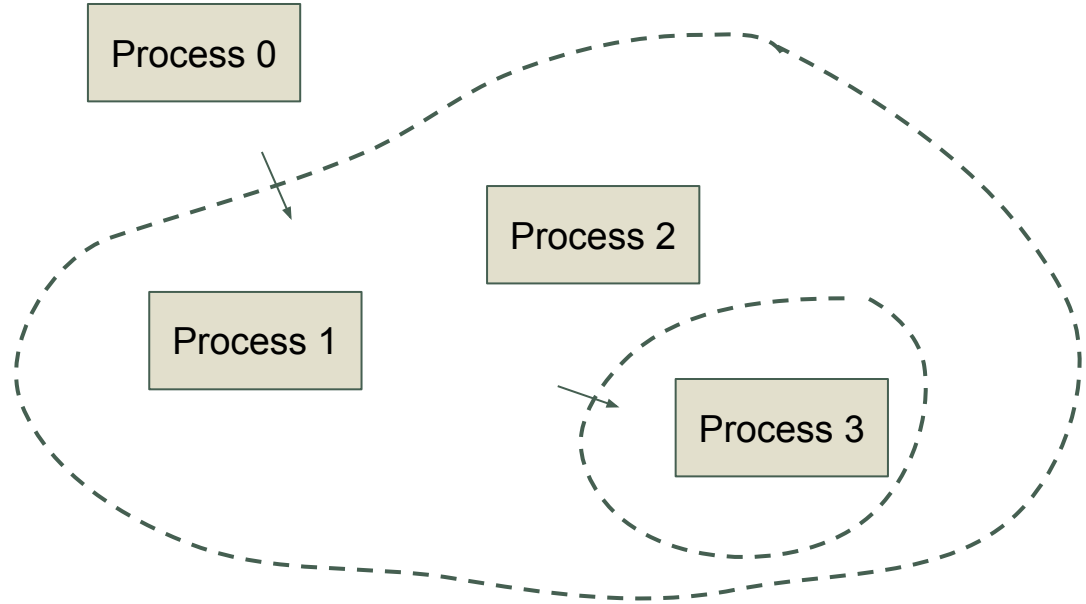
Overview

ABI	Linux	Feature
v6	6.12	Scoped access rights: Abstract Unix Sockets, Signals (Tahera Fatimi)
v7	6.15	Audit Logging (Mickaël Salaün)
v8	7.0	Process-wide ruleset enforcement (TSYNC) (Günther Noack)
v9	7.1	Named Unix Domain Sockets (Günther Noack, Justin Suess, Tingmao Wang...)
v10	7.2	UDP Port Control (Matthieu Buffet)
		Audit Quiet flag (Justin Suess)
v11	7.3	Tracepoints (Mickaël Salaün)
		LANDLOCK_RESTRICT_SELF_NO_NEW_PRIVS (Justin Suess)

Scoped access rights: Abstract Unix Sockets, Signals

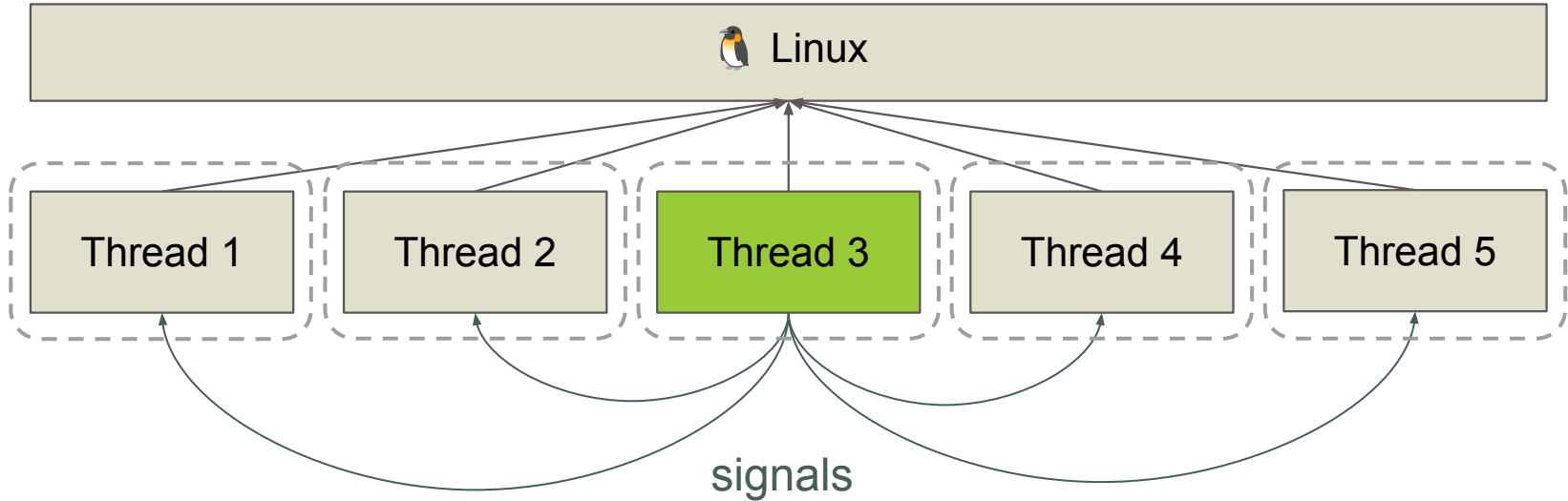
Signals can only cross domain boundaries in one direction (parent to nested).

Same for Abstract Unix Socket connections.



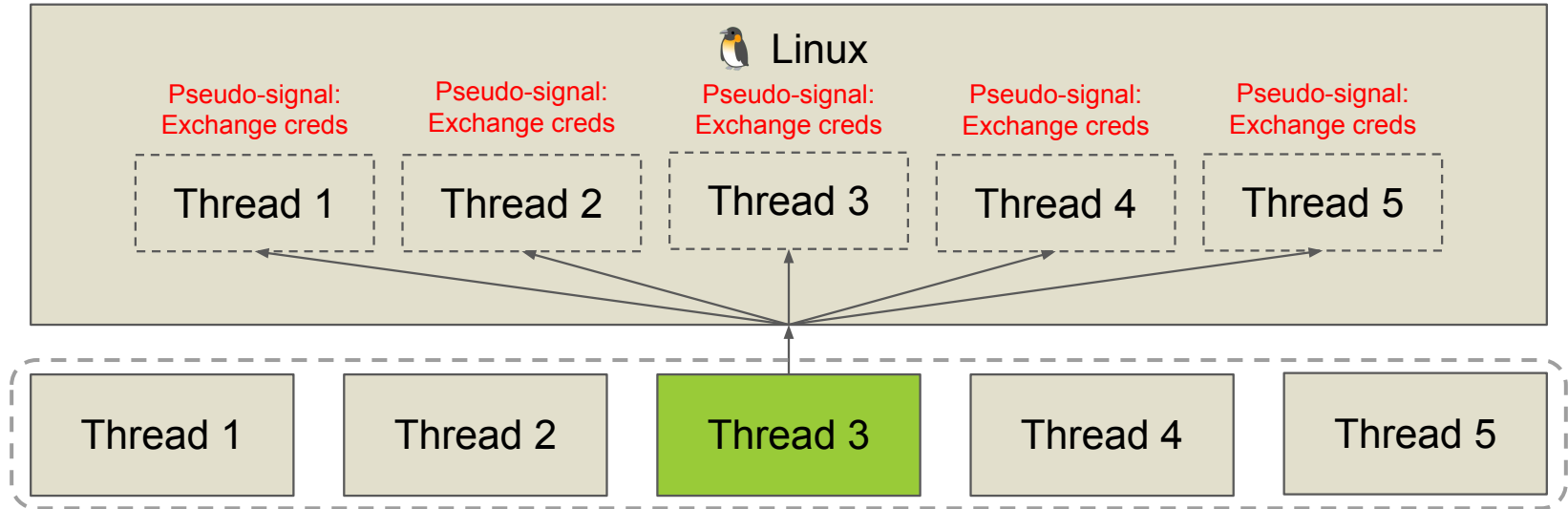
Built by Tahera Fahimi

Process-wide ruleset enforcement (Before)



Tricky userspace workarounds: Signal handlers that enforce Landlock domains. **Libpsx** abstracts this, but it still creates separate Landlock domains.

Process-wide ruleset enforcement (Now :))



Pass `LANDLOCK_RESTRICT_SELF_TSYNC` to `landlock_restrict_self()`
⇒ Implementation difficulty has moved into the Linux kernel

landlock_restrict_self with NO_NEW_PRIVS flag

```
landlock_restrict_self(  
    ruleset_fd, LANDLOCK_RESTRICT_SELF_NO_NEW_PRIVS)
```

Implicit `prctl(SET_NO_NEW_PRIVS)`, simplifying common usage.

Built by Justin Suess

Named Unix Domain Socket resolution

`LANDLOCK_ACCESS_FS_RESOLVE_UNIX` controls connecting and sending messages (for datagram sockets) to Named Unix Domain Sockets (socket files)

- **File system access right:** connections can be allowed by path
- Has "**scoped**" semantics: connections to sockets within the same Landlock domain are always allowed.

Important on environments that run services (e.g. Linux desktop session).

Built in collaboration with Justin Suess, Tingmao Wang, Mickaël Salaün and many others

UDP Port connect() / bind()

Because UDP is not connected, it was more complex (than TCP) to provide a consistent and actionable set of access rights:

- `LANDLOCK_ACCESS_NET_BIND_UDP`
- `LANDLOCK_ACCESS_NET_CONNECT_SEND_UDP`

Control `bind()`, `connect()` and `send()` on UDP ports.

Built by Matthieu Buffet

Audit Logging 1/2

Goals:

- Log unexpected/relevant denials
- Provide enough context to identify the **denied request** and the requester
- Provide enough context to identify the **sandbox creator**

Non goals:

- Dump security policies
- Flood logs with all denials

Audit Logging 2/2

By **default**, only denials related to requests from the same code that enforced a Landlock domain are logged: the enforcer should never try to bypass its own security policy.

User space flags:

- LANDLOCK_RESTRICT_SELF_LOG_SAME_EXEC_OFF
- LANDLOCK_RESTRICT_SELF_LOG_NEW_EXEC_ON
- LANDLOCK_RESTRICT_SELF_LOG_SUBDOMAINS_OFF

Quieting audit logs

New flags in `struct landlock_ruleset_attr` to indicate what access rights should be suppressed for the purpose of audit logging.

```
landlock_add_rule(  
    ruleset_fd, &rule, LANDLOCK_ADD_RULE_QUIET);
```

Suppresses audit logging for the objects (paths, ports) referred to by the rule.

Long term idea: Expand to use in a supervisor mechanism.

Built by Tingmao Wang

Tracepoints

New observability feature, useful to debugging, monitoring, metrics...

An important difference with audit logs is that tracepoints are **disabled by default**.

Because they can be enable one by one and with tailored filters, the Landlock trace events can provide more useful information and context.

Can be used through tracefs and BPF programs, which makes it possible to provide rich and typed context per event.

Landlock events enable us to get **Landlock's state**: domain lifecycle, ruleset's content, denied requests.

Errata

```
landlock_create_ruleset(  
    NULL, 0, LANDLOCK_CREATE_RULESET_ERRATA)
```

returns a bitmask of fixed errata.

- An erratum is a **visible fix** (i.e. behavior change for a corner case) **backported** to all stable kernels, with documentation, and without impact on the Landlock ABI version.
- Detection interface is available on all stable kernels.
- Most programs **should not need to check** these.

Notable bug fixes

- Fix handling of disconnected directories (Mickaël Salaün, Tingmao Wang)
- Always allow signals between threads of the same process (Mickaël Salaün)
- Fix signal scoping bypass on SIGIO path (Bryam Vargas)
- TCP Fastopen connection bypass (Mathieu Buffet)
- Restrict creation of whiteout objects (Günther Noack)
- Fix UAF of source's parent directory (Norbert Szetai)
- Clarify TTY signal scoping (Christopher Lusk)

Thank you all for reporting and fixing bugs!

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Upcoming Features

Namespaces and Capabilities

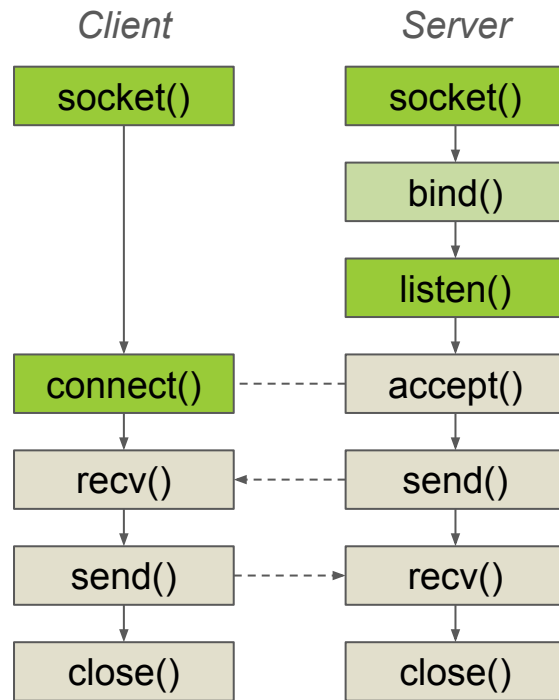
- User namespaces unlock access to kernel code paths which have traditionally been reserved for privileged users.
- New kind of Landlock right to control the use of namespaces (e.g. creation, entering) and the (related) use of capabilities.

Sockets

- Control `socket()` creation by address family, socket type (and protocol)
- Control `listen()` in addition to `bind()` and `connect()`

Better protocol coverage

- Multipath-TCP, ...



Miscellaneous

- Filesystem access
 - Reading and writing file attributes (Cai Xincheng)
- IPC
 - System V Message Queues? (Justin Suess)
 - POSIX Message Queues? (Oxana Kharitonova)
- BPF Interface for applying Landlock rulesets (Justin Suess)
- Threat Model documentation (Mickaël Salaün)
- File system rule deny-listing (no inherit) (Justin Suess)
- Mutable domains and supervisor processes (Tingmao Wang)

Questions



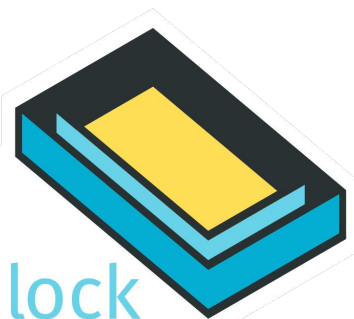
Website: <https://landlock.io/>



Landlock mailing list: <https://subspace.kernel.org/lists.linux.dev.html>



Bug tracker: <https://github.com/landlock-lsm/linux/issues>



Landlock



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